

# Package ‘manMetaVAR’

November 4, 2025

**Title** Multivariate Meta-Analysis of Vector Autoregressive Model Coefficients

**Version** 0.9.4

**Description** Research compendium for the manuscript  
Pesigan, I. J. A., et al. (In Preparation).  
Multivariate Meta-Analysis of Vector Autoregressive Model Coefficients.  
<doi:10.0000/0000000000>.

**URL** <https://github.com/jeksterslab/manMetaVAR>,  
<https://jeksterslab.github.io/manMetaVAR/>,  
<https://osf.io/uenr7/>, <https://doi.org/10.0000/0000000000>

**BugReports** <https://github.com/jeksterslab/manMetaVAR/issues>

**License** MIT + file LICENSE

**Encoding** UTF-8

**LazyData** true

**Roxygen** list(markdown = TRUE)

**Depends** R (>= 4.0.0), OpenMx, fitDTVARMxID, metaVAR

**Imports** stats, utils, simStateSpace, mlVAR

**Remotes** jeksterslab/fitDTVARMxID, jeksterslab/metaVAR

**Suggests** knitr, rmarkdown, testthat

**RoxygenNote** 7.3.3.9000

**NeedsCompilation** no

**Author** Ivan Jacob Agaloos Pesigan [aut, cre, cph] (ORCID:  
<<https://orcid.org/0000-0003-4818-8420>>)

**Maintainer** Ivan Jacob Agaloos Pesigan <r.jeksterslab@gmail.com>

## Contents

|                                  |   |
|----------------------------------|---|
| coef.manmetavar.dtvvar . . . . . | 2 |
| Compress . . . . .               | 4 |

|                                      |    |
|--------------------------------------|----|
| confint.manmetavar.metavar . . . . . | 5  |
| Dynamics . . . . .                   | 6  |
| FitDTVAR . . . . .                   | 6  |
| FitMetaVAR . . . . .                 | 7  |
| FitMLVAR . . . . .                   | 8  |
| FitMplus . . . . .                   | 9  |
| GenData . . . . .                    | 10 |
| MplusInput . . . . .                 | 10 |
| params . . . . .                     | 12 |
| plot.manmetavar.data . . . . .       | 12 |
| print.manmetavar.data . . . . .      | 13 |
| Sim . . . . .                        | 15 |
| SimFitDTVAR . . . . .                | 16 |
| SimFitMetaVAR . . . . .              | 17 |
| SimFitMLVAR . . . . .                | 18 |
| SimFitMplus . . . . .                | 19 |
| SimFN . . . . .                      | 20 |
| SimGenData . . . . .                 | 20 |
| SimProj . . . . .                    | 21 |
| Sum . . . . .                        | 22 |
| SumFitMetaVARLB . . . . .            | 22 |
| SumFitMetaVARNormal . . . . .        | 23 |
| SumFitMLVAR . . . . .                | 24 |
| SumFitMplus . . . . .                | 25 |
| summary.manmetavar.data . . . . .    | 25 |
| vcov.manmetavar.dtvvar . . . . .     | 27 |

## Index 30

---

coef.manmetavar.dtvvar *Parameter Estimates (FitDTVAR)*

---

### Description

Parameter Estimates (FitDTVAR)  
 Parameter Estimates (FitMetaVAR)  
 Parameter Estimates (FitMplus)

### Usage

```
## S3 method for class 'manmetavar.dtvvar'
coef(
  object,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
```

```

    theta = TRUE,
    converged = TRUE,
    grad_tol = 0.01,
    hess_tol = 1e-08,
    vanishing_theta = TRUE,
    theta_tol = 0.001,
    ...
)

## S3 method for class 'manmetavar.metavar'
coef(object, ...)

## S3 method for class 'manmetavar.mplus'
coef(object, median = TRUE, burnin = NULL, ...)
```

### Arguments

|                 |                                                                                                                                                                     |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| object          | Object of class <code>manmetavar.mplus</code> .                                                                                                                     |
| alpha           | Logical. If <code>alpha = TRUE</code> , include estimates of the alpha vector, if available. If <code>alpha = FALSE</code> , exclude estimates of the alpha vector. |
| beta            | Logical. If <code>beta = TRUE</code> , include estimates of the beta matrix, if available. If <code>beta = FALSE</code> , exclude estimates of the beta matrix.     |
| nu              | Logical. If <code>nu = TRUE</code> , include estimates of the nu vector, if available. If <code>nu = FALSE</code> , exclude estimates of the nu vector.             |
| psi             | Logical. If <code>psi = TRUE</code> , include estimates of the psi matrix, if available. If <code>psi = FALSE</code> , exclude estimates of the psi matrix.         |
| theta           | Logical. If <code>theta = TRUE</code> , include estimates of the theta matrix, if available. If <code>theta = FALSE</code> , exclude estimates of the theta matrix. |
| converged       | Logical. Only include converged cases.                                                                                                                              |
| grad_tol        | Numeric scalar. Tolerance for the maximum absolute gradient if <code>converged = TRUE</code> .                                                                      |
| hess_tol        | Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if <code>converged = TRUE</code> .                          |
| vanishing_theta | Logical. Test for measurement error variance going to zero if <code>converged = TRUE</code> .                                                                       |
| theta_tol       | Numeric. Tolerance for vanishing theta test if <code>converged</code> and <code>theta_tol</code> are <code>TRUE</code> .                                            |
| ...             | additional arguments.                                                                                                                                               |
| median          | Logical. If <code>median = TRUE</code> , return median of the posterior. If <code>median = FALSE</code> , return mean of the posterior.                             |
| burnin          | Integer indicating initial samples to discard. If <code>burnin = NULL</code> , do not discard anything.                                                             |

**Value**

Returns a list of vectors of parameter estimates.

Returns a vector of estimated parameters.

Returns a vector of estimated parameters.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

Compress

*Compress Replication*

---

**Description**

Compress Replication

**Usage**

```
Compress(taskid, repid, output_folder)
```

**Arguments**

|               |                                   |
|---------------|-----------------------------------|
| taskid        | Positive integer. Task ID.        |
| repid         | Positive integer. Replication ID. |
| output_folder | Character string. Output folder.  |

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

 confint.manmetavar.metavar

*Confidence Intervals for the Parameter Estimates (FitMetaVAR)*


---

## Description

Confidence Intervals for the Parameter Estimates (FitMetaVAR)

Confidence Intervals for the Parameter Estimates (FitMplus)

## Usage

```
## S3 method for class 'manmetavar.metavar'
confint(object, parm = NULL, level = 0.95, lb = TRUE, ...)

## S3 method for class 'manmetavar.mplus'
confint(object, parm = NULL, level = 0.95, burnin = NULL, ...)
```

## Arguments

|        |                                                                                                                                                                       |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| object | Object of class manmetavar.mplus.                                                                                                                                     |
| parm   | a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered. |
| level  | the confidence level required.                                                                                                                                        |
| lb     | Logical. If TRUE, returns profile likelihood-based confidence intervals. If FALSE, returns Wald confidence intervals.                                                 |
| ...    | additional arguments.                                                                                                                                                 |
| burnin | Integer indicating initial samples to discard. If burnin = NULL, do not discard anything.                                                                             |

## Value

Returns a matrix of confidence intervals.

Returns a matrix of confidence intervals.

## Author(s)

Ivan Jacob Agaloos Pesigan

---

|          |                         |
|----------|-------------------------|
| Dynamics | <i>Process Dynamics</i> |
|----------|-------------------------|

---

**Description**

Process Dynamics

**Usage**

Dynamics(dynamics)

**Arguments**

dynamics            1, 2, or 3. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.

**See Also**

Other Data Generation Functions: [GenData\(\)](#)

**Examples**

```
# stable reciprocal regulation,
Dynamics(dynamics = 1)
# escalating co-activation
Dynamics(dynamics = 2)
# adaptive recovery
Dynamics(dynamics = 3)
```

---

|          |                                                     |
|----------|-----------------------------------------------------|
| FitDTVAR | <i>Fit the Model using the fitDTVARMxID Package</i> |
|----------|-----------------------------------------------------|

---

**Description**

The function fits the model using the [fitDTVARMxID::fitDTVARMxID](#) package.

**Usage**

FitDTVAR(data, ncores = NULL, seed = NULL)

**Arguments**

data                R object. Output of the [GenData\(\)](#) function.  
ncores              Positive integer. Number of cores to use.  
seed                Integer. Random seed.

**See Also**

Other Model Fitting Functions: [FitMLVAR\(\)](#), [FitMetaVAR\(\)](#), [FitMplus\(\)](#), [MplusInput\(\)](#)

**Examples**

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitDTVAR(
  data = data,
  ncores = parallel::detectCores()
)
print(fit)
summary(fit)
coef(fit)
vcov(fit)

## End(Not run)
```

---

FitMetaVAR

*Multivariate Meta-Analysis using the metaVAR Package*

---

**Description**

The function performs multivariate meta-analysis using the [metaVAR::metaVAR](#) package.

**Usage**

```
FitMetaVAR(fit, ncores = NULL)
```

**Arguments**

|                     |                                                              |
|---------------------|--------------------------------------------------------------|
| <code>fit</code>    | R object. Output of the <a href="#">FitDTVAR()</a> function. |
| <code>ncores</code> | Positive integer. Number of cores to use.                    |

**See Also**

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMLVAR\(\)](#), [FitMplus\(\)](#), [MplusInput\(\)](#)

**Examples**

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitDTVAR(
  data = data,
  ncores = parallel::detectCores()
)
```

```

pooled <- FitMetaVAR(
  fit = fit,
  ncores = parallel::detectCores()
)
summary(pooled)
print(pooled)
coef(pooled)
vcov(pooled)

## End(Not run)

```

---

FitMLVAR

*Fit the Model using the mlVAR Package*


---

## Description

The function fits the model using the [mlVAR::mlVAR](#) package.

## Usage

```
FitMLVAR(data, ncores = NULL)
```

## Arguments

|        |                                                             |
|--------|-------------------------------------------------------------|
| data   | R object. Output of the <a href="#">GenData()</a> function. |
| ncores | Positive integer. Number of cores to use.                   |

## See Also

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMetaVAR\(\)](#), [FitMplus\(\)](#), [MplusInput\(\)](#)

## Examples

```

## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitMLVAR(data = data)
print(fit)
summary(fit)

## End(Not run)

```



FitMplus

*Fit the Model using Mplus***Description**

The function fits the model using Mplus.

**Usage**

```
FitMplus(
  data,
  chains = 2L,
  iter = 120000L,
  fscores = NULL,
  plot = FALSE,
  default_priors = TRUE,
  wd = ".",
  mplus_bin = NULL,
  ncores = NULL
)
```

**Arguments**

|                             |                                                                                                                                  |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <code>data</code>           | R object. Output of the <a href="#">GenData()</a> function.                                                                      |
| <code>chains</code>         | Integer. Number of chains.                                                                                                       |
| <code>iter</code>           | Integer. Number of iterations.                                                                                                   |
| <code>fscores</code>        | Integer. Number of iterations for factor scores.                                                                                 |
| <code>plot</code>           | Logical. If <code>plot = TRUE</code> , add <code>PLOT: TYPE = PLOT3;</code> to Mplus input file.                                 |
| <code>default_priors</code> | Logical. If <code>default_priors = TRUE</code> , use default priors.                                                             |
| <code>wd</code>             | Character string. Working directory.                                                                                             |
| <code>mplus_bin</code>      | Character string. Path to Mplus binary. If <code>mplus_bin = NULL</code> , the function will try to find the appropriate binary. |
| <code>ncores</code>         | Positive integer. Number of cores to use.                                                                                        |

**See Also**

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMLVAR\(\)](#), [FitMetaVAR\(\)](#), [MplusInput\(\)](#)

**Examples**

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitMplus(data = data)
print(fit)
```

```
summary(fit)

## End(Not run)
```

---

|         |                      |
|---------|----------------------|
| GenData | <i>Simulate Data</i> |
|---------|----------------------|

---

**Description**

The function simulates data using the `simStateSpace::SimSSMIVary()` function.

**Usage**

```
GenData(taskid)
```

**Arguments**

taskid                    Positive integer. Task ID.

**See Also**

Other Data Generation Functions: [Dynamics\(\)](#)

**Examples**

```
set.seed(42)
data <- GenData(taskid = 1)
print(data)
summary(data)
plot(data)
```

---

|            |                                |
|------------|--------------------------------|
| MplusInput | <i>Create Mplus Input File</i> |
|------------|--------------------------------|

---

**Description**

The function creates an Mplus input file.

**Usage**

```
MplusInput(
  dynamics,
  fn_data,
  fn_posterior,
  fn_factorscores,
  chains,
  iter,
  fscores,
  plot,
  default_priors,
  ncores = NULL
)
```

**Arguments**

|                 |                                                                                                              |
|-----------------|--------------------------------------------------------------------------------------------------------------|
| dynamics        | 1, 2, or 3. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery. |
| fn_data         | Character string. Filename for data file.                                                                    |
| fn_posterior    | Character string. Filename for posterior output.                                                             |
| fn_factorscores | Character string. Filename for factor scores output.                                                         |
| chains          | Integer. Number of chains.                                                                                   |
| iter            | Integer. Number of iterations.                                                                               |
| fscores         | Integer. Number of iterations for factor scores.                                                             |
| plot            | Logical. If plot = TRUE, add PLOT: TYPE = PLOT3; to Mplus input file.                                        |
| default_priors  | Logical. If default_priors = TRUE, use default priors.                                                       |
| ncores          | Positive integer. Number of cores to use.                                                                    |

**See Also**

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMLVAR\(\)](#), [FitMetaVAR\(\)](#), [FitMplus\(\)](#)

**Examples**

```
cat(
  MplusInput(
    dynamics = 1,
    fn_data = "data.dat",
    fn_posterior = "posterior.dat",
    fn_factorscores = "factorscores.dat",
    chains = 2L,
    iter = 60000L,
    fscores = 1000L,
    plot = TRUE,
    default_priors = FALSE,
    ncores = NULL
  )
)
```

```
)  
)
```

---

|        |                              |
|--------|------------------------------|
| params | <i>Simulation Parameters</i> |
|--------|------------------------------|

---

**Description**

Simulation Parameters

**Usage**

params

**Format**

A dataframe with 25 rows and 3 columns:

- taskid** Simulation Task ID.
- n** Sample size.
- time** Number of measurement occassions.
- dynamics** Process dynamics. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|                                   |                                                           |
|-----------------------------------|-----------------------------------------------------------|
| <code>plot.manmetavar.data</code> | <i>Plot Method for an Object of Class manmetavar.data</i> |
|-----------------------------------|-----------------------------------------------------------|

---

**Description**

- Plot Method for an Object of Class `manmetavar.data`
- Plot Method for an Object of Class `manmetavar.mplus`

**Usage**

```
## S3 method for class 'manmetavar.data'
plot(x, ...)

## S3 method for class 'manmetavar.mplus'
plot(
  x,
  what = "posterior",
  parm = NULL,
  level = 0.95,
  burnin = NULL,
  legend_loc = "topright",
  ...
)
```

**Arguments**

|            |                                                                                                                                                                       |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| x          | Object of class <code>manmetavar.mplus</code> .                                                                                                                       |
| ...        | additional arguments.                                                                                                                                                 |
| what       | Character string. If <code>what = "posterior"</code> , return posterior distribution plots. If <code>what = "trace"</code> , return trace plots.                      |
| parm       | a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered. |
| level      | the confidence level required.                                                                                                                                        |
| burnin     | Integer indicating initial samples to discard. If <code>burnin = NULL</code> , do not discard anything.                                                               |
| legend_loc | Character string. Legend location.                                                                                                                                    |

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

`print.manmetavar.data` *Print Method for an Object of Class `manmetavar.data`*

---

**Description**

Print Method for an Object of Class `manmetavar.data`  
 Print Method (FitDTVAR)  
 Print Method (FitMetaVAR)  
 Print Method (FitMLVAR)

**Usage**

```
## S3 method for class 'manmetavar.data'
print(x, ...)

## S3 method for class 'manmetavar.dtvar'
print(
  x,
  means = FALSE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  digits = 4,
  ...
)

## S3 method for class 'manmetavar.metavar'
print(x, alpha = 0.05, digits = 4, ...)

## S3 method for class 'manmetavar.mlvar'
print(
  x,
  show = c("fit", "temporal", "contemporaneous", "between"),
  digits = 4,
  ...
)
```

**Arguments**

|                    |                                                                                                                                                                 |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>x</code>     | Object of class <code>manmetavar.mlvar</code> .                                                                                                                 |
| <code>...</code>   | additional arguments.                                                                                                                                           |
| <code>means</code> | Logical. If <code>means = TRUE</code> , return means. Otherwise, the function returns raw estimates.                                                            |
| <code>alpha</code> | Numeric vector. Significance level $\alpha$ .                                                                                                                   |
| <code>beta</code>  | Logical. If <code>beta = TRUE</code> , include estimates of the beta matrix, if available. If <code>beta = FALSE</code> , exclude estimates of the beta matrix. |
| <code>nu</code>    | Logical. If <code>nu = TRUE</code> , include estimates of the nu vector, if available. If <code>nu = FALSE</code> , exclude estimates of the nu vector.         |
| <code>psi</code>   | Logical. If <code>psi = TRUE</code> , include estimates of the psi matrix, if available. If <code>psi = FALSE</code> , exclude estimates of the psi matrix.     |

|                 |                                                                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| theta           | Logical. If theta = TRUE, include estimates of the theta matrix, if available. If theta = FALSE, exclude estimates of the theta matrix. |
| converged       | Logical. Only include converged cases.                                                                                                  |
| grad_tol        | Numeric scalar. Tolerance for the maximum absolute gradient if converged = TRUE.                                                        |
| hess_tol        | Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if converged = TRUE.            |
| vanishing_theta | Logical. Test for measurement error variance going to zero if converged = TRUE.                                                         |
| theta_tol       | Numeric. Tolerance for vanishing theta test if converged and theta_tol are TRUE.                                                        |
| digits          | Integer indicating the number of decimal places to display.                                                                             |
| show            | Tables to show.                                                                                                                         |

**Value**

Returns a matrix of estimates, standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

Sim

---

*Simulation Replication*


---

**Description**

Simulation Replication

**Usage**

```
Sim(
  taskid,
  repid,
  output_folder,
  overwrite,
  integrity,
  seed,
  data,
  metavar,
  mlvar,
  mplus,
  chains,
```

```
    iter,  
    fscores,  
    plot,  
    default_priors  
  )
```

**Arguments**

|                |                                                                                           |
|----------------|-------------------------------------------------------------------------------------------|
| taskid         | Positive integer. Task ID.                                                                |
| repid          | Positive integer. Replication ID.                                                         |
| output_folder  | Character string. Output folder.                                                          |
| overwrite      | Logical. Overwrite existing output in output_folder.                                      |
| integrity      | Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE. |
| seed           | Integer. Random seed.                                                                     |
| data           | Logical. Simulate data.                                                                   |
| metavar        | Logical. If metavar = TRUE, fit the model using the metaVAR package.                      |
| mlvar          | Logical. If mlvar = TRUE, fit the model using the mlVAR package.                          |
| mplus          | Logical. If mplus = TRUE, fit the model using DSEM in Mplus.                              |
| chains         | Integer. Number of chains.                                                                |
| iter           | Integer. Number of iterations.                                                            |
| fscores        | Integer. Number of iterations for factor scores.                                          |
| plot           | Logical. If plot = TRUE, add PLOT: TYPE = PLOT3; to Mplus input file.                     |
| default_priors | Logical. If default_priors = TRUE, use default priors.                                    |

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|             |                                          |
|-------------|------------------------------------------|
| SimFitDTVAR | <i>Simulation Replication - FitDTVAR</i> |
|-------------|------------------------------------------|

---

**Description**

Simulation Replication - FitDTVAR

**Usage**

```
SimFitDTVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```



**Arguments**

|               |                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                                            |
| repid         | Positive integer. Replication ID.                                                                                     |
| output_folder | Character string. Output folder.                                                                                      |
| seed          | Integer. Random seed.                                                                                                 |
| suffix        | Character string. Output of <code>manCTMed:::SimSuffix()</code> .                                                     |
| overwrite     | Logical. Overwrite existing output in <code>output_folder</code> .                                                    |
| integrity     | Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> . |

**Details**

This function is executed via the `Sim` function.

**Value**

The output is saved as an external file in `output_folder`.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

SimFitMetaVAR

*Simulation Replication - FitMetaVAR*


---

**Description**

Simulation Replication - FitMetaVAR

**Usage**

```
SimFitMetaVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

**Arguments**

|               |                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                                            |
| repid         | Positive integer. Replication ID.                                                                                     |
| output_folder | Character string. Output folder.                                                                                      |
| seed          | Integer. Random seed.                                                                                                 |
| suffix        | Character string. Output of <code>manCTMed:::SimSuffix()</code> .                                                     |
| overwrite     | Logical. Overwrite existing output in <code>output_folder</code> .                                                    |
| integrity     | Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> . |

**Details**

This function is executed via the Sim function.

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|             |                                          |
|-------------|------------------------------------------|
| SimFitMLVAR | <i>Simulation Replication - FitMLVAR</i> |
|-------------|------------------------------------------|

---

**Description**

Simulation Replication - FitMLVAR

**Usage**

SimFitMLVAR(taskid, repid, output\_folder, seed, suffix, overwrite, integrity)

**Arguments**

|               |                                                                                           |
|---------------|-------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                |
| repid         | Positive integer. Replication ID.                                                         |
| output_folder | Character string. Output folder.                                                          |
| seed          | Integer. Random seed.                                                                     |
| suffix        | Character string. Output of manCTMed:::SimSuffix().                                       |
| overwrite     | Logical. Overwrite existing output in output_folder.                                      |
| integrity     | Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE. |

**Details**

This function is executed via the Sim function.

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

SimFitMplus

*Simulation Replication - FitMplus***Description**

Simulation Replication - FitMplus

**Usage**

```

SimFitMplus(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  chains,
  iter,
  fscores,
  plot,
  default_priors
)

```

**Arguments**

|                |                                                                                                                       |
|----------------|-----------------------------------------------------------------------------------------------------------------------|
| taskid         | Positive integer. Task ID.                                                                                            |
| repid          | Positive integer. Replication ID.                                                                                     |
| output_folder  | Character string. Output folder.                                                                                      |
| seed           | Integer. Random seed.                                                                                                 |
| suffix         | Character string. Output of <code>manCTMed:::SimSuffix()</code> .                                                     |
| overwrite      | Logical. Overwrite existing output in <code>output_folder</code> .                                                    |
| integrity      | Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> . |
| chains         | Integer. Number of chains.                                                                                            |
| iter           | Integer. Number of iterations.                                                                                        |
| fscores        | Integer. Number of iterations for factor scores.                                                                      |
| plot           | Logical. If <code>plot = TRUE</code> , add <code>PLOT: TYPE = PLOT3</code> ; to Mplus input file.                     |
| default_priors | Logical. If <code>default_priors = TRUE</code> , use default priors.                                                  |

**Details**

This function is executed via the `Sim` function.

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|       |                             |
|-------|-----------------------------|
| SimFN | <i>Simulation File Name</i> |
|-------|-----------------------------|

---

**Description**

Simulation File Name

**Usage**

SimFN(output\_type, output\_folder, suffix)

**Arguments**

- output\_type      Character string. Output type. Valid values include "data", "fit-dt-var-mx", "fit-meta-var-mx", and "fit-ml-var".
- output\_folder    Character string. Output folder.
- suffix            Character string. Output of manCTMed::SimSuffix().

**Value**

Returns a character string file name with the output\_folder in the OS-specific format.

---

|            |                                         |
|------------|-----------------------------------------|
| SimGenData | <i>Simulation Replication - GenData</i> |
|------------|-----------------------------------------|

---

**Description**

Simulation Replication - GenData

**Usage**

SimGenData(taskid, repid, output\_folder, seed, suffix, overwrite, integrity)

**Arguments**

|               |                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                                            |
| repid         | Positive integer. Replication ID.                                                                                     |
| output_folder | Character string. Output folder.                                                                                      |
| seed          | Integer. Random seed.                                                                                                 |
| suffix        | Character string. Output of <code>manCTMed:::SimSuffix()</code> .                                                     |
| overwrite     | Logical. Overwrite existing output in <code>output_folder</code> .                                                    |
| integrity     | Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> . |

**Details**

This function is executed via the `Sim` function.

**Value**

The output is saved as an external file in `output_folder`.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|         |                                |
|---------|--------------------------------|
| SimProj | <i>Simulation Project Name</i> |
|---------|--------------------------------|

---

**Description**

Simulation Project Name

**Usage**

```
SimProj()
```

**Value**

Returns the project name as a character string.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|     |                |
|-----|----------------|
| Sum | <i>Summary</i> |
|-----|----------------|

---

**Description**

Summary

**Usage**

```
Sum(taskid, reps, output_folder, overwrite, integrity, metavar, mlvar, mplus)
```

**Arguments**

|               |                                                                                           |
|---------------|-------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                |
| reps          | Positive integer. Number of replications.                                                 |
| output_folder | Character string. Output folder.                                                          |
| overwrite     | Logical. Overwrite existing output in output_folder.                                      |
| integrity     | Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE. |
| metavar       | Logical. If metavar = TRUE, fit the model using the metaVAR package.                      |
| mlvar         | Logical. If mlvar = TRUE, fit the model using the mlVAR package.                          |
| mplus         | Logical. If mplus = TRUE, fit the model using DSEM in Mplus.                              |

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|                 |                                               |
|-----------------|-----------------------------------------------|
| SumFitMetaVARLB | <i>Summary: Likelihood-Based (FitMetaVAR)</i> |
|-----------------|-----------------------------------------------|

---

**Description**

Summary: Likelihood-Based (FitMetaVAR)

**Usage**

```
SumFitMetaVARLB(taskid, reps, output_folder, overwrite, integrity)
```

**Arguments**

|               |                                                                                           |
|---------------|-------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                |
| reps          | Positive integer. Number of replications.                                                 |
| output_folder | Character string. Output folder.                                                          |
| overwrite     | Logical. Overwrite existing output in output_folder.                                      |
| integrity     | Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE. |

**Details**

This function is executed via the Sum function.

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|                     |                                            |
|---------------------|--------------------------------------------|
| SumFitMetaVARNormal | <i>Summary: Normal Theory (FitMetaVAR)</i> |
|---------------------|--------------------------------------------|

---

**Description**

Summary: Normal Theory (FitMetaVAR)

**Usage**

```
SumFitMetaVARNormal(taskid, reps, output_folder, overwrite, integrity)
```

**Arguments**

|               |                                                                                           |
|---------------|-------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                |
| reps          | Positive integer. Number of replications.                                                 |
| output_folder | Character string. Output folder.                                                          |
| overwrite     | Logical. Overwrite existing output in output_folder.                                      |
| integrity     | Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE. |

**Details**

This function is executed via the Sum function.

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|             |                           |
|-------------|---------------------------|
| SumFitMLVAR | <i>Summary (FitMLVAR)</i> |
|-------------|---------------------------|

---

**Description**

Summary (FitMLVAR)

**Usage**

SumFitMLVAR(taskid, reps, output\_folder, overwrite, integrity)

**Arguments**

|               |                                                                                           |
|---------------|-------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                |
| reps          | Positive integer. Number of replications.                                                 |
| output_folder | Character string. Output folder.                                                          |
| overwrite     | Logical. Overwrite existing output in output_folder.                                      |
| integrity     | Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE. |

**Details**

This function is executed via the Sum function.

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan



---

|             |                           |
|-------------|---------------------------|
| SumFitMplus | <i>Summary (FitMplus)</i> |
|-------------|---------------------------|

---

**Description**

Summary (FitMplus)

**Usage**

SumFitMplus(taskid, reps, output\_folder, overwrite, integrity)

**Arguments**

- |               |                                                                                           |
|---------------|-------------------------------------------------------------------------------------------|
| taskid        | Positive integer. Task ID.                                                                |
| reps          | Positive integer. Number of replications.                                                 |
| output_folder | Character string. Output folder.                                                          |
| overwrite     | Logical. Overwrite existing output in output_folder.                                      |
| integrity     | Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE. |

**Details**

This function is executed via the Sum function.

**Value**

The output is saved as an external file in output\_folder.

**Author(s)**

Ivan Jacob Agaloos Pesigan

---

|                         |                                                              |
|-------------------------|--------------------------------------------------------------|
| summary.manmetavar.data | <i>Summary Method for an Object of Class manmetavar.data</i> |
|-------------------------|--------------------------------------------------------------|

---

**Description**

- Summary Method for an Object of Class manmetavar.data
- Summary Method (FitDTVAR)
- Summary Method (FitMetaVAR)
- Summary Method (FitMLVAR)
- Summary Method (FitMplus)

**Usage**

```
## S3 method for class 'manmetavar.data'
summary(object, ...)

## S3 method for class 'manmetavar.dtvvar'
summary(
  object,
  means = FALSE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  digits = 4,
  ...
)

## S3 method for class 'manmetavar.metavar'
summary(object, alpha = 0.05, digits = 4, ...)

## S3 method for class 'manmetavar.mlvar'
summary(
  object,
  show = c("fit", "temporal", "contemporaneous", "between"),
  digits = 4,
  ...
)

## S3 method for class 'manmetavar.mplus'
summary(object, alpha = 0.05, median = TRUE, digits = 4, burnin = NULL, ...)
```

**Arguments**

|        |                                                                                                                                                                 |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| object | Object of class <code>manmetavar.mplus</code> .                                                                                                                 |
| ...    | additional arguments.                                                                                                                                           |
| means  | Logical. If <code>means = TRUE</code> , return means. Otherwise, the function returns raw estimates.                                                            |
| alpha  | Numeric vector. Significance level $\alpha$ .                                                                                                                   |
| beta   | Logical. If <code>beta = TRUE</code> , include estimates of the beta matrix, if available. If <code>beta = FALSE</code> , exclude estimates of the beta matrix. |
| nu     | Logical. If <code>nu = TRUE</code> , include estimates of the nu vector, if available. If <code>nu = FALSE</code> , exclude estimates of the nu vector.         |

|                 |                                                                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| psi             | Logical. If psi = TRUE, include estimates of the psi matrix, if available. If psi = FALSE, exclude estimates of the psi matrix.         |
| theta           | Logical. If theta = TRUE, include estimates of the theta matrix, if available. If theta = FALSE, exclude estimates of the theta matrix. |
| converged       | Logical. Only include converged cases.                                                                                                  |
| grad_tol        | Numeric scalar. Tolerance for the maximum absolute gradient if converged = TRUE.                                                        |
| hess_tol        | Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if converged = TRUE.            |
| vanishing_theta | Logical. Test for measurement error variance going to zero if converged = TRUE.                                                         |
| theta_tol       | Numeric. Tolerance for vanishing theta test if converged and theta_tol are TRUE.                                                        |
| digits          | Integer indicating the number of decimal places to display.                                                                             |
| show            | Tables to show.                                                                                                                         |
| median          | Logical. If median = TRUE, return median of the posterior. If median = FALSE, return mean of the posterior.                             |
| burnin          | Integer indicating initial samples to discard. If burnin = NULL, do not discard anything.                                               |

### Value

Returns a matrix of estimates, standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

Returns a matrix of estimates, standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

Returns a matrix of estimates, standard errors, number of posterior samples, and confidence intervals.

### Author(s)

Ivan Jacob Agaloos Pesigan

---

vcov.manmetavar.dtvvar *Sampling Covariance Matrix of the Parameter Estimates (FitDTVAR)*

---

### Description

Sampling Covariance Matrix of the Parameter Estimates (FitDTVAR)

Sampling Covariance Matrix of the Parameter Estimates (FitMetaVAR)

Sampling Covariance Matrix of the Parameter Estimates (FitMplus)

**Usage**

```
## S3 method for class 'manmetavar.dtvvar'
vcov(
  object,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  ...
)

## S3 method for class 'manmetavar.metavar'
vcov(object, ...)

## S3 method for class 'manmetavar.mplus'
vcov(object, burnin = NULL, ...)
```

**Arguments**

|                              |                                                                                                                                                                     |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>object</code>          | Object of class <code>manmetavar.mplus</code> .                                                                                                                     |
| <code>alpha</code>           | Logical. If <code>alpha = TRUE</code> , include estimates of the alpha vector, if available. If <code>alpha = FALSE</code> , exclude estimates of the alpha vector. |
| <code>beta</code>            | Logical. If <code>beta = TRUE</code> , include estimates of the beta matrix, if available. If <code>beta = FALSE</code> , exclude estimates of the beta matrix.     |
| <code>nu</code>              | Logical. If <code>nu = TRUE</code> , include estimates of the nu vector, if available. If <code>nu = FALSE</code> , exclude estimates of the nu vector.             |
| <code>psi</code>             | Logical. If <code>psi = TRUE</code> , include estimates of the psi matrix, if available. If <code>psi = FALSE</code> , exclude estimates of the psi matrix.         |
| <code>theta</code>           | Logical. If <code>theta = TRUE</code> , include estimates of the theta matrix, if available. If <code>theta = FALSE</code> , exclude estimates of the theta matrix. |
| <code>converged</code>       | Logical. Only include converged cases.                                                                                                                              |
| <code>grad_tol</code>        | Numeric scalar. Tolerance for the maximum absolute gradient if <code>converged = TRUE</code> .                                                                      |
| <code>hess_tol</code>        | Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if <code>converged = TRUE</code> .                          |
| <code>vanishing_theta</code> | Logical. Test for measurement error variance going to zero if <code>converged = TRUE</code> .                                                                       |
| <code>theta_tol</code>       | Numeric. Tolerance for vanishing theta test if <code>converged</code> and <code>theta_tol</code> are <code>TRUE</code> .                                            |

|        |                                                                                           |
|--------|-------------------------------------------------------------------------------------------|
| ...    | additional arguments.                                                                     |
| burnin | Integer indicating initial samples to discard. If burnin = NULL, do not discard anything. |

**Value**

Returns a list of sampling variance-covariance matrices.

Returns the sampling variance-covariance matrix of the estimated parameters.

Returns the sampling variance-covariance matrix of the estimated parameters.

**Author(s)**

Ivan Jacob Agaloos Pesigan

# Index

- \* **Compression Functions**
  - Compress, [4](#)
- \* **Data Generation Functions**
  - Dynamics, [6](#)
  - GenData, [10](#)
- \* **Model Fitting Functions**
  - FitDTVAR, [6](#)
  - FitMetaVAR, [7](#)
  - FitMLVAR, [8](#)
  - FitMplus, [9](#)
  - MplusInput, [10](#)
- \* **compress**
  - Compress, [4](#)
- \* **data**
  - params, [12](#)
- \* **fit**
  - FitDTVAR, [6](#)
  - FitMLVAR, [8](#)
  - FitMplus, [9](#)
  - MplusInput, [10](#)
  - SimFitDTVAR, [16](#)
  - SimFitMetaVAR, [17](#)
  - SimFitMLVAR, [18](#)
  - SimFitMplus, [19](#)
- \* **gendata**
  - Dynamics, [6](#)
  - GenData, [10](#)
  - SimGenData, [20](#)
- \* **manCTMed**
  - SimFN, [20](#)
- \* **manMetaVAR**
  - Compress, [4](#)
  - Dynamics, [6](#)
  - FitDTVAR, [6](#)
  - FitMetaVAR, [7](#)
  - FitMLVAR, [8](#)
  - FitMplus, [9](#)
  - GenData, [10](#)
  - MplusInput, [10](#)

- Sim, [15](#)
- SimFitDTVAR, [16](#)
- SimFitMetaVAR, [17](#)
- SimFitMLVAR, [18](#)
- SimFitMplus, [19](#)
- SimGenData, [20](#)
- SimProj, [21](#)
- Sum, [22](#)
- SumFitMetaVARLB, [22](#)
- SumFitMetaVARNormal, [23](#)
- SumFitMLVAR, [24](#)
- SumFitMplus, [25](#)
- \* **meta**
  - FitMetaVAR, [7](#)
- \* **methods**
  - coef.manmetavar.dtvvar, [2](#)
  - confint.manmetavar.metavar, [5](#)
  - plot.manmetavar.data, [12](#)
  - print.manmetavar.data, [13](#)
  - summary.manmetavar.data, [25](#)
  - vcov.manmetavar.dtvvar, [27](#)
- \* **parameters**
  - params, [12](#)
- \* **simulation**
  - Sim, [15](#)
  - SimFitDTVAR, [16](#)
  - SimFitMetaVAR, [17](#)
  - SimFitMLVAR, [18](#)
  - SimFitMplus, [19](#)
  - SimFN, [20](#)
  - SimGenData, [20](#)
  - SimProj, [21](#)
  - Sum, [22](#)
  - SumFitMetaVARLB, [22](#)
  - SumFitMetaVARNormal, [23](#)
  - SumFitMLVAR, [24](#)
  - SumFitMplus, [25](#)
- \* **summary**
  - Sum, [22](#)

- SumFitMetaVARLB, [22](#)
- SumFitMetaVARNormal, [23](#)
- SumFitMLVAR, [24](#)
- SumFitMplus, [25](#)
- coef.manmetavar.dtvvar, [2](#)
- coef.manmetavar.metavar
  - (coef.manmetavar.dtvvar), [2](#)
- coef.manmetavar.mplus
  - (coef.manmetavar.dtvvar), [2](#)
- Compress, [4](#)
- confint.manmetavar.metavar, [5](#)
- confint.manmetavar.mplus
  - (confint.manmetavar.metavar), [5](#)
- Dynamics, [6](#), [10](#)
- FitDTVAR, [6](#), [7–9](#), [11](#)
- FitDTVAR(), [7](#)
- fitDTVARMxID::fitDTVARMxID, [6](#)
- FitMetaVAR, [7](#), [7–9](#), [11](#)
- FitMLVAR, [7](#), [8](#), [9](#), [11](#)
- FitMplus, [7](#), [8](#), [9](#), [11](#)
- GenData, [6](#), [10](#)
- GenData(), [6](#), [8](#), [9](#)
- metaVAR::metaVAR, [7](#)
- mlVAR::mlVAR, [8](#)
- MplusInput, [7–9](#), [10](#)
- params, [12](#)
- plot.manmetavar.data, [12](#)
- plot.manmetavar.mplus
  - (plot.manmetavar.data), [12](#)
- print.manmetavar.data, [13](#)
- print.manmetavar.dtvvar
  - (print.manmetavar.data), [13](#)
- print.manmetavar.metavar
  - (print.manmetavar.data), [13](#)
- print.manmetavar.mlvar
  - (print.manmetavar.data), [13](#)
- Sim, [15](#)
- SimFitDTVAR, [16](#)
- SimFitMetaVAR, [17](#)
- SimFitMLVAR, [18](#)
- SimFitMplus, [19](#)
- SimFN, [20](#)
- SimGenData, [20](#)
- SimProj, [21](#)
- simStateSpace::SimSSMIVary(), [10](#)
- Sum, [22](#)
- SumFitMetaVARLB, [22](#)
- SumFitMetaVARNormal, [23](#)
- SumFitMLVAR, [24](#)
- SumFitMplus, [25](#)
- summary.manmetavar.data, [25](#)
- summary.manmetavar.dtvvar
  - (summary.manmetavar.data), [25](#)
- summary.manmetavar.metavar
  - (summary.manmetavar.data), [25](#)
- summary.manmetavar.mlvar
  - (summary.manmetavar.data), [25](#)
- summary.manmetavar.mplus
  - (summary.manmetavar.data), [25](#)
- vcov.manmetavar.dtvvar, [27](#)
- vcov.manmetavar.metavar
  - (vcov.manmetavar.dtvvar), [27](#)
- vcov.manmetavar.mplus
  - (vcov.manmetavar.dtvvar), [27](#)